

Report

To investigate the correlation between copy number variations in the *CNSL* region and ethnicity, we analyzed NGS read depths across probes. While median read depths of nonCNSL regions vary slightly between samples, no outliers with extremely low coverage are observed (**Fig. 1**). Following the normalization steps described by Fromer and Purcell (2014), four probes (CNSL_probe_5, CNSL_probe_23, CNSL_probe_28, and CNSL_probe_46) still have high variability (standard deviation > 0.5) across all samples (**Fig. 2**), likely due to poor specificity or cross-hybridization. These probes, for which normalization likely failed, were excluded from further analysis and may require redesign.

Analysis of deletion and duplication frequencies at four breakpoints reveal distinct ethnicity-specific patterns (**Table 1, Fig. 3**). Breakpoints 1 and 2 occur exclusively in ethnicity group A, with deletion and duplication frequencies of ~1.1% and ~1.2%, respectively. Breakpoint 3 are specific to ethnicity group B, with deletion and duplication frequencies of ~1.0% and ~5.2%, respectively. Lastly, breakpoint 4 is found exclusively in ethnicity group C, with deletion and duplication frequencies of ~0.57% and ~2.8%, respectively.

A Bayesian probabilistic approach can be employed to determine the ethnicity of unlabeled samples. First, deletion and duplication frequencies at breakpoints in the unlabeled samples are characterized and calculated. Using Bayes' theorem, the ethnicity with the highest posterior probability is identified based on the likelihood of the observed frequencies and the prior prevalence of each ethnic group. This approach is formalized by the following equation:

$$P(E|F) = P(F|E) * P(E) / P(F)$$

$P(E|F)$ = Posterior probability of ethnicity E given the observed del/dup frequencies F

$P(F|E)$ = Likelihood of observing del/dup frequencies F for a specific ethnicity E

$P(E)$ = Prior probability of the ethnicity E (based on population prevalence)

$P(F)$ = Probability of observing del/dup frequencies F across all ethnicities.

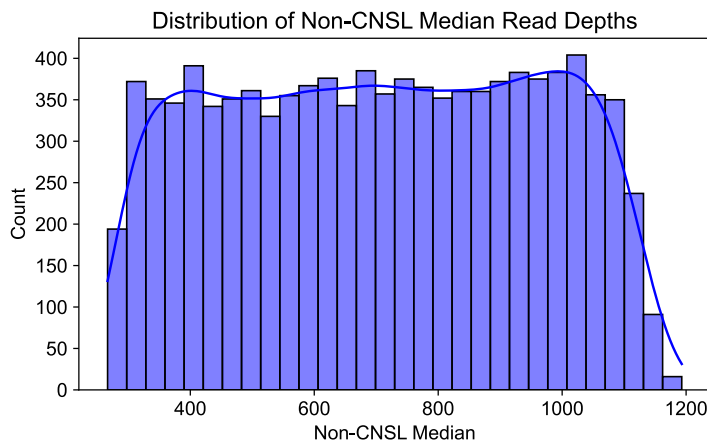


Figure 1. Distribution of median read depths of samples in the non-CNSL regions.

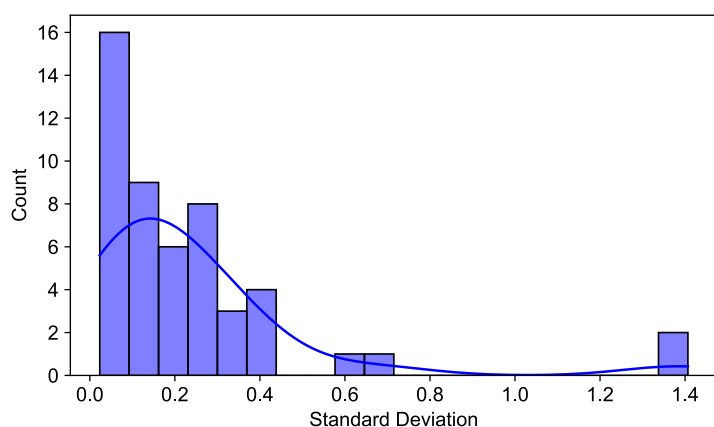


Figure 2. Distribution of standard deviations across all CNSL probes. Four probes (CNSL_probe_5, CNSL_probe_23, CNSL_probe_28, and CNSL_probe_46) have high variability (standard deviation > 0.5).

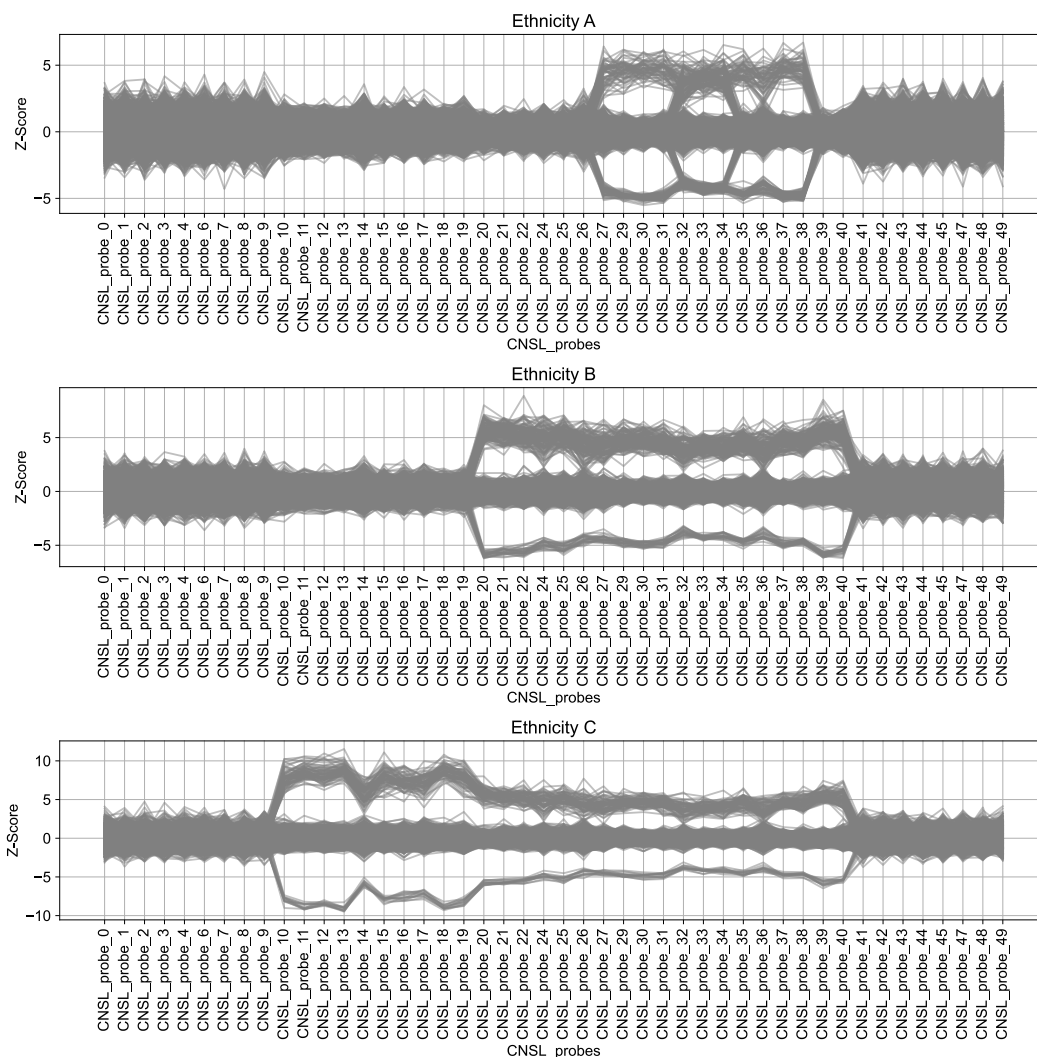


Figure 3. Z-scores of read depth for selected probes across different ethnic groups. Problematic probes are excluded.

Table 1: Deletion and Duplication Frequencies at CNSL Breakpoints by Ethnic Group (A, B and C)

Del/Dup index	5' breakpoint	3' breakpoint	Ethnicity A (del/dup)	Ethnicity B (del/dup)	Ethnicity C (del/dup)	All (del/dup)
1	CNSL probe 32	CNSL probe 38	1.1%/1.2%	0.0%/0.0%	0.0%/0.0%	0.57%/0.62%
2	CNSL probe 27	CNSL probe 34	1.1%/1.2%	0.0%/0.0%	0.0%/0.0%	0.55%/0.61%
3	CNSL probe 20	CNSL probe 40	0.0%/0.0%	1.0%/5.2%	0.0%/0.0%	0.26%/1.32%
4	CNSL probe 10	CNSL probe 40	0.0%/0.0%	0.0%/0.0%	0.6%/2.8%	0.14%/0.69%

Reference

Fromer, Menachem, and Shaun M. Purcell. 2014. "UsingXHMM Software to Detect Copy Number Variation in Whole-Exome Sequencing Data." *Et Al [Current Protocols in Human Genetics]* 81 (1): 7.23.1–7.23.21.